SIMATS ENGINEERING

ASSESSMENT TOOL 1: DATA INTERPRETATION

Course Name: Computer Vision
Course Code: ITA05
Course Outcome: CO4 – Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics.
Assessment Tool Weightage: 40%

Question 8: Model Generalization vs Overfitting

Given Data

Model	Training Accuracy	Testing Accuracy
Model A	98%	70%
Model B	90%	88%

Comparison and Analysis

Model A achieves a very high training accuracy of 98%, but its testing accuracy decreases significantly to 70%. This large difference indicates that Model A has likely overfitted the training data. It performs very well on known data but does not generalize effectively to unseen data.

Model B has a training accuracy of 90% and a testing accuracy of 88%. The small difference between training and testing accuracy indicates better generalization. Although its training accuracy is lower than Model A, it performs much more consistently on unseen data.

Evaluation

For deployment, Model B is more reliable because real-world systems must perform well on new and unseen inputs. Model A has a higher training accuracy, but its large testing-performance drop creates a higher risk of failure after deployment.

Conclusion

Model B should be selected for deployment. Its smaller training-testing gap shows better generalization, lower overfitting risk, and more reliable real-world performance.

Question 11: Detection Model Comparison under Resource Constraints

Given Data

Model	**Accuracy**	**Latency**
Model A	85%	40 ms
Model B	92%	90 ms
Model C	90%	60 ms

Comparison and Analysis

Model A has the lowest latency of 40 ms, making it the fastest model, but its accuracy is only 85%.

Model B provides the highest accuracy of 92%, but it also has the highest latency of 90 ms. This may be unsuitable for applications requiring faster responses.

Model C provides 90% accuracy with 60 ms latency. It offers a good balance between detection accuracy and processing speed.

Evaluation

For a real-time system with moderate accuracy requirements, Model C is the most suitable. Model A can be selected when speed is the highest priority, while Model B is preferable when accuracy is more important than response time.

Conclusion

Model C is the optimal balanced choice because it provides high accuracy while maintaining reasonably low latency. It gives a better trade-off between performance and real-time responsiveness.

Question 15: GMM Performance in Different Conditions

Given Data

Scenario	**Accuracy**
Static Background	92%
Dynamic Background	68%

Comparison and Analysis

The GMM-based system achieves 92% accuracy in a static background. This indicates that GMM performs well when the background remains relatively stable.

However, accuracy decreases to only 68% in a dynamic background. Dynamic elements such as moving trees, shadows, water, or changing illumination can be incorrectly classified as foreground objects.

The accuracy difference is:

92% − 68% = 24 percentage points

This significant reduction demonstrates that GMM is sensitive to background changes.

Evaluation

GMM alone is not sufficiently reliable for all environments. It is suitable for controlled environments with relatively stable backgrounds, but its performance can decrease considerably in dynamic scenes.

Suggested Improvements

The system can be improved by:

1. Using adaptive background updating.
2. Applying noise-reduction and morphological operations.
3. Using shadow removal techniques.
4. Combining GMM with optical flow.
5. Using modern object detection methods such as YOLO for better object discrimination.

Conclusion

GMM is effective for static environments but insufficient by itself for highly dynamic environments. Combining GMM with additional motion or object-detection techniques can improve robustness.

Question 18: Detection Model Generalization

Given Data

Model	**Training Accuracy**	**Testing Accuracy**
Model A	97%	75%
Model B	90%	88%

Comparison and Analysis

Model A achieves 97% training accuracy, but its testing accuracy is only 75%. The difference is:

$97\% - 75\% = 22$ percentage points

This large gap suggests that Model A has overfitted the training data.

Model B has 90% training accuracy and 88% testing accuracy, giving a difference of only:

$90\% - 88\% = 2$ percentage points

This small gap indicates that Model B generalizes much better to unseen data.

Evaluation

Model A may appear better because it has higher training accuracy, but its low testing accuracy makes it less reliable for deployment.

Model B provides slightly lower training accuracy but much better testing performance. Therefore, it is more likely to maintain its performance on real-world inputs.

Conclusion

Model B is better for deployment because it has stronger generalization, a much smaller training-testing gap, and a lower risk of overfitting.

Question 30: System Performance under Constraints

Given Data

Model	**Accuracy**	**Latency**	**Cost**
Model A	88%	50 ms	Low
Model B	92%	80 ms	Moderate
Model C	95%	120 ms	High

Comparison and Analysis

Model A has the lowest accuracy of 88%, but it provides the lowest latency of 50 ms and has a low cost. Therefore, it is suitable for applications where speed and affordability are important.

Model B provides 92% accuracy with 80 ms latency and moderate cost. It provides a good balance between accuracy, speed, and cost.

Model C provides the highest accuracy of 95%, but it has the highest latency of 120 ms and high cost. Therefore, it may not be suitable for systems with strict response-time and budget constraints.

Evaluation

Considering all three factors, Model B provides the best overall balance. Model A is preferable when low cost and fast response are the highest priorities. Model C should be selected when maximum accuracy is more important than latency and cost.

Conclusion

Model B is the best overall choice because it provides high accuracy while maintaining moderate latency and cost. It offers a practical trade-off between accuracy, speed, and resource requirements.

Overall Conclusion

The comparisons show that selecting a computer vision model should not depend only on accuracy. Latency, computational requirements, robustness, generalization, and cost must also be considered.

A model with the highest accuracy may not always be the best choice for real-world deployment. The most suitable technique is the one that provides an appropriate balance between performance and application requirements.

Code of Conduct

I ___O.S.Sabiha(192424264)___________________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: __Sabiha____________________